element14
AN AVNET COMMUNITY

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Raspberry Pi and the Importance of Urban and Indoor Farming

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Raspberry Pi and the Importance of Urban and Indoor Farming

element14 is a Community of over 750,000 makers, professional engineers, electronics enthusiasts, and everyone in between. Since our beginnings in 2009, we have provided a place to discuss electronics, get help with your designs and projects, show off your skills by building a new prototype, and much more. We also offer online learning courses such as our Essentials series, video tutorials from element14 Presents, and electronics competitions with our Design Challenges.

Large scale agriculture is both essential to humanity's future, and a significant contributor to the greenhouse gases that are changing our climate. In this eBook, we will discuss potential alternative agriculture methods like indoor and urban farming, how the Raspberry Pi computer can be a powerful tool for those techniques, and we will also examine some example applications of innovative growing techniques. We hope this eBook might spur you to try out your own DIY agriculture setup using the Raspberry Pi, and also check out our other eBook series for free learning on many other topics.

element14 Community Team

CHAPTER - 1 GREENHOUSE GAS EMISSIONS

According to a 2020 report from the Food and Agriculture Organization of the United Nations, the world population currently stands at 7.8 billion people, and is expected to grow by 25% by 2050. That's a lot of mouths to feed, and to meet that demand, farmers will have to increase their global food production by 50%, a feat much easier said than done. Would you be surprised to learn that farmers worldwide already produce enough food to feed 10 billion people? That same FAO report outlines that food production has increased faster than the global population growth, yet millions of people still starve from a lack of nourishment.

A large chunk of the problem can be blamed on poverty, with people in some countries having little to no income being unable to afford to eat or feed their families. That problem alone can be further compounded by several factors, including defunct or fully absent infrastructure preventing food delivery, leaving aside the issue of then having it processed for consumption. If that wasn't bad enough, food security and the nutritional status of those in the vulnerable population groups have deteriorated further due to the health and socio-economic impacts of the COVID-19 pandemic. The world is currently trying to mitigate the fallout from the pandemic, and as a result, seasonal farmworkers have been restricted from travel, consumers have altered their shopping habits, some production facilities have closed, and tariffs have impacted food trade policies.

The percentage of greenhouse gas emissions emitted by the transportation, energy, and agriculture industries

There is also the issue of greenhouse gas (GHG) emissions and their contribution to climate change via agriculture practices, which can be attributed to land management and clearing, accounting for about 1/5 of total gases released into the atmosphere. That's more than all of the world's vehicles, planes, ships, and trains combined, according

to Environment Reports. Emissions from farming and deforestation also contribute more gas than the global building industry, with only the energy sector producing more, by about 37%. Agricultural management GHGs can be broken down into three components, including carbon dioxide (CO2) from tropical deforestation, methane (CH4) from livestock and rice production, and nitrous oxide (N2O) from fertilizing or burning croplands.

Livestock, including cows and sheep, produce a significant amount of methane through feeding, and rice production also produces a significant amount (around 11% of GHGs) of methane via biomass cultivation (decomposition). Adding to those numbers is field cultivation, such as burning biomass and the use of fertilizer (manure) to enrich the soil, which is caused by microbes feeding on the nitrogen, transforming it into nitrous oxide which is then released into the atmosphere.

Of course, all of the food we cultivate and harvest doesn't get to grocery store shelves all on its own. It has to be transported using trucks, planes, trains, and ships, which themselves add to the GHG problem, as food products usually have to travel long distances to reach distributors, and subsequently, grocery chains and farmer's markets. For example, in the US during the winter season, most of our produce is grown and shipped from the lower states, Mexico, and South America. 'Food Miles' is a term, I'll wager, most of us have never heard; it equates to the distance food travels before being consumed, which includes being transported to a processing center/ distributor and then heading to local retailers.

To that end, it's estimated that food travels an average of 1,500 miles in the US before being consumed at home, which translates into almost 10 kcal of fossil fuel energy burned for our food system for every 1 kcal of energy we derive from said food. If we look at GHG emissions as a whole, aircraft, ships, trains, and trucks account for about 6% of produced GHGs, which isn't bad, but it could still use a reduction in its carbon footprint.

All of those GHGs in the agriculture industry are a poor tradeoff when it comes to food production and the environment. Sure, we can enjoy Big Macs and Mocha Latte Cappuccinos now, but our grandkids' children will be stuck with generic, flavorless Soylent Green down the line if we don't change how we grow our food using sustainable practices and new technologies. While it may sound as if we are headed for a future that looks like a barren wasteland, there are many ways to mitigate those greenhouse gas emissions, including using urban and indoor farming using our friend, the **Raspberry Pi**.

Food transportation to distribution centers and grocery outlets is another major contributor to GHGs

CHAPTER - 2 URBAN & INDOOR FARMING

Urban and indoor farming can lessen GHGs by growing, processing, and distributing food in local areas

Urban and indoor farming is the practice of cultivating (growing/nurturing), processing, and distributing food in or around urban areas. In other words, it's growing and distributing food locally with minimal impact on the environment. There is no "one solution fits all" when it comes to urban farming, meaning there is no particular method for producing sustainable crops and animal liveries. It's more of what solution best fits the local communities and environments. For example, inner cities may benefit more from using vertical hydroponic gardening, while some suburban areas could best utilize automated indoor farming.

Urban and indoor farming aren't new concepts; there's evidence that ancient Sumerians and Persians created oases near settlements, using aqueducts that carried water from mountain runoffs and waste from local communities. Urban farming has since been used during times of economic downfall, with local communities using abandoned lots to grow food during the Great Depression, and during times of war with Victory Gardens.

With that said, there are several types of urban farming practices that are being widely adopted as sustainability and necessity continue to develop. These ventures can help enhance local soils, reduce carbon emissions, and improve air quality.

1: Community Supported Agriculture

One of the more popular urban farming endeavors is community-supported agriculture (CSA), which aims to keep produce fresh and local, relying on local food producers to serve a set number of urban members who agree to pay a subscription fee for a share of the grown produce. CSAs can reduce shipping costs and the GHGs produced by limiting the transportation of goods over long distances. This type of urban farming can also boost communities' economic development via farmers markets that sell leftover produce.

2: Vertical Farming

Vertical farming is a means of food production that grows food vertically instead of on a single, horizontal plane, as conventional or greenhouse farming may employ. This type may be used in limited space areas, such as the inside of buildings, shipping containers, or even in abandoned mine shafts. It's usually combined with innovative growing techniques such as hydroponics or aquaponics and situated in climate-controlled environments. The significant benefit to using this type of farming is to produce more food in a limited space, especially in urban environments where space is at a premium. Vertical farming is commonly part of skyscrapers or repurposed warehouses within a city.

3: Hydroponics/Aquaponics/Aeroponics

Hydroponics is the growing of plants without soil; instead, plants are immersed in nutrient-rich water (or water is run over the roots) with gravel acting as a support structure. The water used in these systems can come from recycled sources and be mixed with chemical fertilizers or organic matter, which saves local resources. Water can be introduced using several methods, including wicks that transport water passively from a reservoir using a capillary material such as cotton rope.

Aquaponics is similar to hydroponics, only in this case, fish and other aquatic life are grown and nurtured. Most often, the fish are grown with plants for a self-contained cycle that nourishes them both. Fish are provided food, and their waste produces ammonia, and bacteria in the water converts that ammonia into nutrients for the plants. A filter helps to keep the cycle flowing, resulting in a beneficial ecosystem.

Aeroponic systems provide nourishment using nutrient-infused water mist, and are similar to hydroponics in that the plants are typically anchored in a soilless medium such as gravel. Roots are generally exposed in this method, where they are 'puffed' by specially-designed misting devices that keep the roots from drying out.

4: Backyard, Rooftop, Forest,and Container

As the title suggests, this type of farming utilizes any extra space found around, on, or in homes and buildings to grow food. Even local forests can be utilized to some degree without damaging local ecosystems. These types of urban farming methods depend on the weather, unless used in conjunction with a climate-controlled shelter, or they can be brought indoors during adverse weather.

Careful consideration should be given to rooftop gardening, as plants will need to be rooted in either a raised bed or in containers, and the weight of the soil can damage the roof if it can't handle the weight. Water becomes another factor using this method, due to ease of access and additional weight constraints. Some rooftop setups are typically watered by hand using waterspouts, while others take advantage of drip systems designed using a system of porous hoses and pumps that allow water to leak through at individual rates.

Each method has its pros and cons, and each should be chosen by which method would be suitable for that particular environment. For example, vertical farming is ideal for areas with space constraints, such as cities, while CSAs thrive in city outskirts and suburban areas. Resources also play a role in which method is best incorporated, such as water availability and the type of lighting needed for both plants and animals.

It is also just as important to note that even self-contained environments can have their problems in terms of plant health. For example, it's not uncommon for plants in vertical farms to be inflicted with certain molds and other pests, due to improper conditions and the growing medium's contamination. This particular method also often relies on experts to set up the growing system, as specific PH levels need to be maintained, specialized lighting has to be implemented, and pollination is problematic at best.

The same goes for hydroponics, aeroponics, and aquaponics, as each needs a specialized environment to

maintain healthy, optimal growth. Entire fish farms have been wiped out due to improper chemical water conditioner use, while plants grown in hydroponic systems have wilted and died due to improper PH levels. There's a reason why Venus Flytrap carnivorous plants only grow in some regions of North and South Carolina, and not in the Rocky Mountains. Nature has evolved to provide the perfect ecosystem for any given plant, and any deviation of climate norms can wipe out even the hardiest of plants and animals.

Beyond strict maintenance regimens, costs are also a drawback to these farming types, as most rely on technology for growth and nourishment, such as expensive lighting, water filtration/conditioning systems, and specialized growing mediums. These methods also have upfront building costs that can be pricey when it comes to materials and labor, as some systems can be complicated and require extensive planning. In addition, there can be operational costs required to maintain optimal plant growth, as these systems rely on energy to power lights and other systems.

CHAPTER - 3 RASPBERRY PI & AGRICULTURE

Raspberry Pi 4 Model B

The last few decades have given rise to more affordable computer technology in the form of SoCs (Systems on a Chip), SiPs (Systems in a Package), SoMs (Systems on a Module), CoMs (Computers on a Module), and other three-letter platforms, which are all forms of a complete (or nearly complete) computer in a tiny form factor. Beyond the processor, SoCs typically feature RAM, GPU, and some form of storage. Those chips form the basis for single-board computers (SBCs), incorporating a myriad of different I/Os, such as Ethernet, HDMI, USB, audio, and more. They also include headers for attaching add-on boards and peripherals that provide different functionalities, such as wireless communications, sensors, relays, cameras, and more.

That same affordable technology has also given birth to the Internet of Things (IoT), a network of physical objects (or things) that are embedded with sensors and other technology (hardware and software), which can connect with other devices and systems via the internet. Those devices can include everything from smart appliances to remote sensing platforms; almost anything that can collect and share data can be part of the IoT ecosystem.

To that end, one of the most popular SBCs on the market is the Raspberry Pi, which was introduced nearly a decade ago and has evolved from a classroom learning tool to a driving force of innovation for engineers and makers alike. This compact board has been used in many different projects, including home automation systems, plotters, retro gaming platforms, and robotics, and has even found its way into NASA's Jet Propulsion Laboratory Open Source Rover.

In its current form, the Raspberry Pi 4 Model B, the board packs a Broadcom BCM2711 quad-core Cortex-A72 (ARM v8) 64-bit SoC with up to 4 GB of LPDDR4-2400 SDRAM (user-specified). It also offers several different connection options, including 2.4 GHz and 5.0 GHz IEEE 802. 11B/g/n/ac Wireless LAN, Bluetooth 5.0, and double-true Gigabit Ethernet. Port options are also extensive and feature 2X USB 3.0 ports, 2X USB 2.0 Ports, and 2X micro-HDMI ports supporting a 4Kp60 video resolution. Additional features include 2-lane MIPI DSI/CSI ports for cameras and displays, 4-pole stereo audio and composite video port, Micro SD card slot for operating systems and data storage, and the standard 40 pin GPIO (General Purpose In/Out) header found on most Pi boards from its release in 2012.

Software, such as Mozilla's Things Gateway, allows users to program their smart home projects using the Raspberry Pi and other SBCs

As mentioned earlier, the GPIO header allows users to connect different add-on boards to increase the Pi's functionality, such as adding sensors, relays, cameras, LEDs, and even audio DACs. The Raspberry Pi Foundation designed the board to be utilized as a stand-alone unit, or also as a foundation that could be modified and upgraded for different applications. For example, the Pi can be tasked as the brain for home automation systems, controlling lighting, televisions/entertainment centers, appliances, thermostats, security cameras, and more.

SBCs, including the Pi, use software to program their projects in one form or another. Some use easy-to-navigate GUIs with drag-and-drop features, such as EZBlock Pi, while others employ extensive programming languages that rely on scripts, such as Python. While both have options for all programming skill levels, more advanced engineers and makers often use script languages. SBCs like the Raspberry Pi can also use operating systems much in the same fashion as PCs, and these are mostly Linux-based, such as Rasbian, Debian, LibreELEC, and even Android.

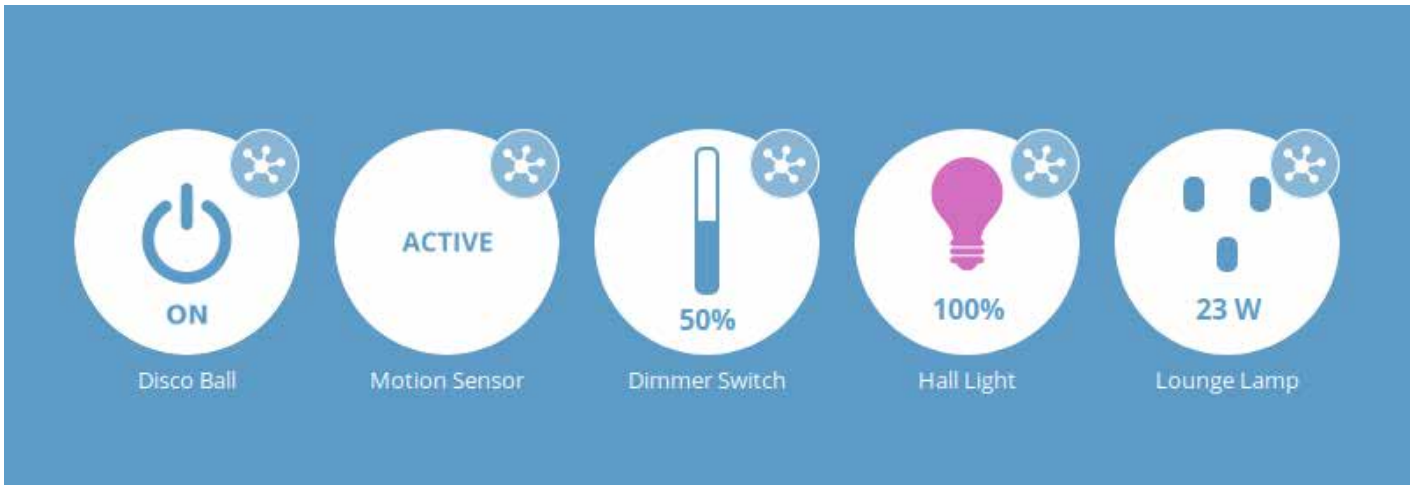


Image Source: Mozilla

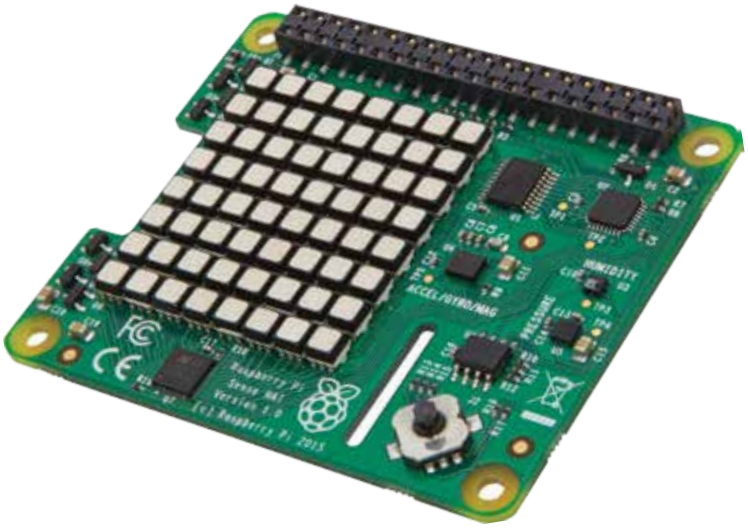
The Raspberry Pi Sense Hat connects to the Pi via a 40 pin header and is outfitted with several different sensors that can monitor pressure, humidity, temperature, and orientation

As the Raspberry Pi has been an excellent foundation for any smart home system over the last decade, urban farmers have taken notice and have been actively using the board for their respective farming projects. The Raspberry Pi has proven to be a reliable, cheap, and easy-to-use platform for a myriad of projects, including urban farming. The Pi is an open-source board, and the Raspberry Pi Foundation has released schematics for the boards, which means its design can be revamped how users see fit. Any number of software systems can be utilized depending on users' requirements, making the Pi an excellent board for any project, including farming.

Another reason the Pi is beneficial for urban farming is that it can be outfitted with sensors and relays to control systems via add-on boards, known as HATs, which connect to the Pi via headers. The Pi can also take advantage of add-on boards from other manufacturers and makers, including Adafruit, which has an extensive Feather and Wing catalog compatible with the Pi. Like the Pi, these add-on boards are designed to be affordable, often costing $20 USD or less, depending on the outlet.

When used in tandem, not only can users implement a reliable setup for urban farming, it can also be done without breaking the bank.

Sensors are a crucial component of any urban farming setup, allowing users to monitor and collect data on a variety of parameters and values, including climate (air temperature, humidity levels, CO2, and airflow rates), plant temperature, water nutrient composition, light levels/spectrum, and plant morphology (deficiencies and growth). Those conditions play a significant role for various reasons. For example, the difference between plant temperature and air temperature can provide information on whether a plant's pores (stomata) are open or closed. If closed, the plant can't absorb CO2 and convert it into biomass. The same can be said for measuring pH levels and conductivity in irrigation water, which are vital for propagation and growth. In short, sensors allow urban farmers to monitor growth conditions, identify anomalies, and correct any problems early on in the growth cycle, providing a proactivity level unseen in traditional farming.



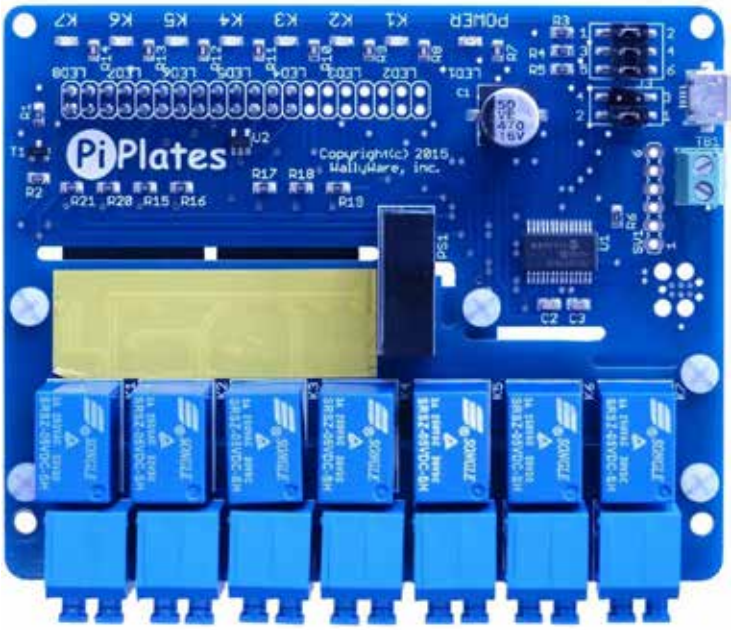
Raspberry Pi Sense Hat

Pi-Plates' Relay Plate can automate various urban farming systems, including irrigation and water systems, lighting, cameras, and more

Relays are another critical piece of hardware for urban farming in terms of automation, which include contactors, reed, latching, and solid-state (among a host of others), with each designed for specific voltages and applications. They are electrically operated switches used to turn various systems on and off without human intervention. In the case of automated farming, relays can control specialized lighting, such as LED and high-intensity ballast lamps that utilize a multitude of different spectrums, including blue (430 to 450 nm, used for seedlings), red (640 to 680 nm, fruit production), and green/yellow (utilized by certain plants for photosynthesis).

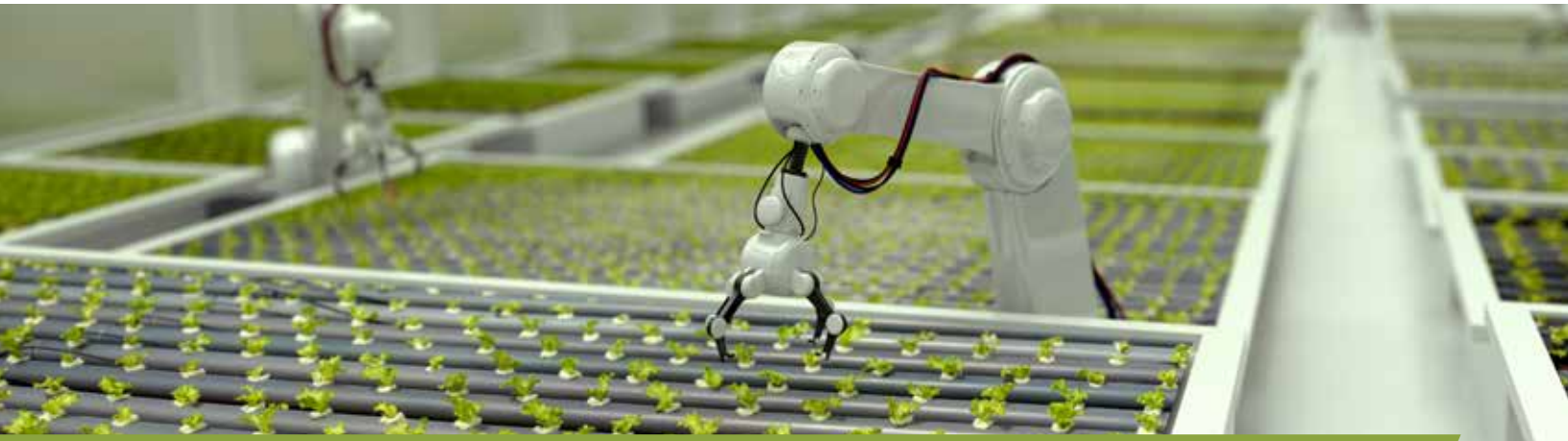
Relays are also used to control water pumps, such as immersion, peristaltic, trash, multistage, centrifugal, and others, depending on the application and the amount of water required for each system. Irrigation systems also rely on relays, which include drip (low-flow water introduction), localized (specific areas), sprinkler (overhead, ground level), surface (topsoil irrigation), subsurface (root irrigation), and a few specialized others. The types of plants being grown will dictate the best irrigation method. For example, corn is most often grown with sprinkler irrigation, as it requires a lot of water, but other farmers have recently found that drip systems can yield the same amount of growth using less water, as only the roots receive water with this setup.

Of course, relays are also used to control fans and heating systems to achieve plants' optimal environmental conditions to thrive. These are most often used in conjunction with temperature and moisture sensors to maintain those environments with minimal or no fluctuations—the same for heating mats, which are designed to keep the root systems of tropical plants and fruits warm.



Pi-Plates' Relay Plate

CHAPTER - 4 AUTOMATION & ENCLOSED ENVIRONMENTS



Enclosed environments with automated systems can provide nominal plant and fruit production, catering to plants' specific needs

All of the information presented in this article is just the tip of the iceberg regarding the amount of knowledge there is on urban farming alone, much less automation. There are college classes designed for any given aspect of plant species, growth, and production, touching on what grow zones (climates) are optimal, how much water is needed, what type of water, what type of light, and which spectrum and soil compositions are best, to consider just a few questions you may have when implementing your own system. Urban farming is both a science and an art, and cultivating plants for optimum growth can be a huge mountain to climb. That said, it's not impossible, and can be a fun endeavor!

As with any project, good foundations begin with questions: how much water is needed, when should I water, and what is the right airflow and heat environment, to take a few examples. Do day and night cycles affect plants, and is there a way to force plants to produce more fruits and vegetables for larger yields? A good rule of thumb is, to begin with, the basics of growing plants and setting the high-tech platforms and systems aside for a brief moment, to get a better understanding of how plants grow and what they require for optimal growth.

In this example, we'll take a look at growing a tomato plant. While there are many different varieties, we'll use a determinate (bush) variety with no particular cultivar and a container for growth instead of the ground. Tomatoes are a warm-weather crop, and do best in temperate climates with a lot of sunlight. Since we're growing tomatoes in containers, we'll use starter plants already grown from seedlings and placed into a pot with composted soil, although a store-bought mix explicitly designed for tomatoes is best.

Recently planted tomatoes will require a lot of sunlight, or, if growing indoors, a temperate UV grow light works well. They also require a lot of water, about 1.2 gallons every week, and thrive in room temperatures (65 to 75 ˚F). Watering should be done in the early morning to give the plant the moisture it needs to thrive during the day, as growth is slowed during evening hours for lack of photosynthesis. Keep the same light schedule if you're using grow lights, to reduce shock and promote healthy growth. Fertilization can be done via a water solution, and should be administered every two weeks when the fruit is about an inch in diameter to propagate healthy growth further.

The tomatoes should be left on the plant for as long as possible and are ripe when the fruit is firm and red in color; some will have a bit of yellow around the stem, but these are still considered ripe. Other varieties have different colors, but determining their ripeness is nearly universal in terms of firmness. This basic example should give you an idea of what's involved with growing plants. Different plants require a different set of growing environments to thrive, but some can be grouped in the same environment if the conditions for each don't deviate too much, such as growing tomatoes, garlic, and squash.

Growing tomatoes can be easy if done correctly, and they can thrive in urban gardens as long as their requirements are met. Now that you have a basic understanding of what's required to grow plants, the process can be automated with the same specific needs they require when cultivated by hand. That said, there are a ton of urban garden projects that use the versatile Pi, and listed below are a few great examples.

Tomatoes and other fruits are easy to grow and can thrive in urban gardens

CHAPTER - 5 PI FARMING PROJECTS

1: Space Vegetables

Feiticeir0's Space Vegetables project uses an interior (greenhouse) hydroponic garden to grow vegetables, as part of element14's 1 Meter of Pi Challenge, which tasked contestants to design a system that can produce the most food in 1m3 of space. Feiticer0 designed the Space Vegetables interior hydroponic garden using plywood and transparent plastic sheeting for the greenhouse, which is outfitted with white, red, and blue LEDs on an overhead railing, which act as grow lights with a broad spectrum.

Inside the tiny greenhouse is a hydroponic kit that features a series of tubes that are filled with a coconut coir-based growing medium with beefsteak tomatoes, beans, and lettuce that were grown from seed. A series of pumps are used to bring nutrient-infused water and air to the plants for optimum growing conditions.

The Space Vegetables project uses a Raspberry Pi 4, Automation Hat, Enviro Hat, and Pico Hat Hacker to

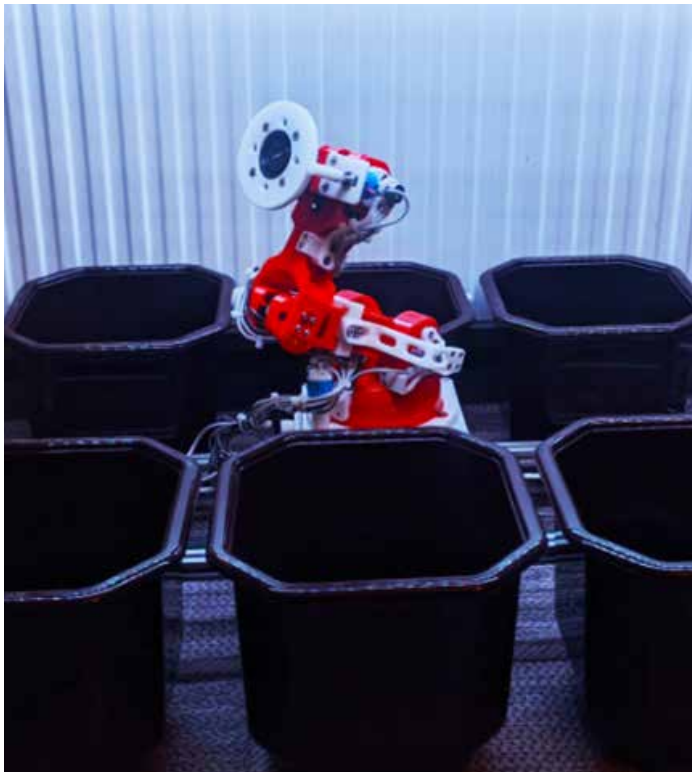
control the system. The Enviro Hat is outfitted with a series of sensors that monitor temperature, pressure, light, and even motion, which Feiticeir0 used for different values (pH, humidity, temperature, etc.) and which are handled by the Pi 4 with access to a database server. A Raspberry Pi 3 B+ controls the automation, while a Pi Zero and Pi camera are used to take daily images and environmental status, which are then posted on Twitter daily for easy and remote monitoring. Feiticeir0 used Flak, Rest, and Web Services to garner the data, take pictures, and upload them to Twitter in an easy-to-read package with useful information.

2: Project RAG

Project RAG (Robot-Assisted Gardening) is another excellent entry for element14's 1 Meter of Pi Challenge. It was designed by Milos Rasic for astronauts traveling to Mars as a way to produce food when on the red planet. Milos designed Project RAG using a robotic arm outfitted with a quick-change head that can be used to swap out a series of different modules, including a sensor and light system, water module, and robotic grippers. The robot rides along a rail system on a motorized base to tend to a series of plants housed in grow buckets along the sides of the track for easy tending.

The entire system resides in a small greenhouse enclosure, and is controlled using a Raspberry Pi 4. Simultaneously, an Arduino Nano handles the sensors and other hardware within the modules, which can be switched out with a novel quick tool change mechanism. The robot rides along the rail and uses its sensors to take various readings for optimal plant growth, and delivers water via a hose to the plants within each bucket when needed. An overhead LED rig supplies the plants with efficient lighting, which can be adjusted for different intensities and spectrums.

Milos created code for the platform from scratch using Python and PyQt5, and uses I2C for communication between the modules, and employs a novel GUI for controlling the rig, lighting, pump, Pi camera, and everything else using a series of on-screen sliders and buttons. Says Milos in a blog post, "This was a really fun project, but not a small one for sure," indicating that some automated farming setups can require a fair amount of time and technical knowledge. The project required mechanical and electrical design, as well as software programming.



3: FarmBot C.N.C. Farming

Image Source: FarmBot

FarmBot is an open-source precision CNC farming system designed by mechanical engineer Rory Aronson to grow crops in limited spaces. The robotic platform can plant and cultivate over 30 different crops, including potatoes, peas, beans, lettuce, artichokes, and more in an area that measures 2.9 X 1.4 meters. What's more, it can cultivate those crops within the same area and operate indoors, outdoors, and within covered areas. It's also estimated that FarmBot produces 25% fewer emissions than standard US food production processes.

The robotic system uses a CNC railing with linear guides and an interchangeable tool head designed to carry out all processes before harvesting, including heads for sowing seeds, weeding, and watering, and even uses a specialized sensor attachment for testing soil. The system uses a belt and pulley arrangement driven by Nema 17 stepper motors with rotary encoders, along with GT2 belts and OpenBuilds' V-Slot extrusions and wheels. Controlling the FarmBot is done using a Raspberry Pi 3, with an Arduino Mega 2560 with a RAMPS 4.1 shield and camera to record data.

The Raspberry Pi uses those sensors, along with a camera and AI, to identify plants and even their age using images and other data on the internet. This also allows it to identify weeds and remove them from the garden area. Moreover, it can take into account local weather conditions and adjust its watering schedule depending on the amount of rain it may receive. The system uses the OpenFarm database to create an optimal planting plan based on crop size, and uses a web-based interface to program and control the FarmBot locally on mobile devices or remotely via the app.

These were just a few examples of farming and gardening projects that can be accomplished using the Raspberry Pi, which can be utilized and scaled for many different systems depending on the applications. Urban farming is continuously evolving using the latest technologies and ever-changing climate conditions. Whether you believe in climate change or not, we can all agree that producing food locally can only benefit those communities who take advantage of it. The Raspberry Pi is just the latest tool that can benefit those gardens, and help those within those communities to produce viable crops annually and cheaply. At the end of the day, it's about feeding those in need, and having some fun while doing so!

For more information about **Raspberry Pi**, check out our dedicated web page. See **Design Challeges** based around the Pi, **Pi videos**, and more.

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